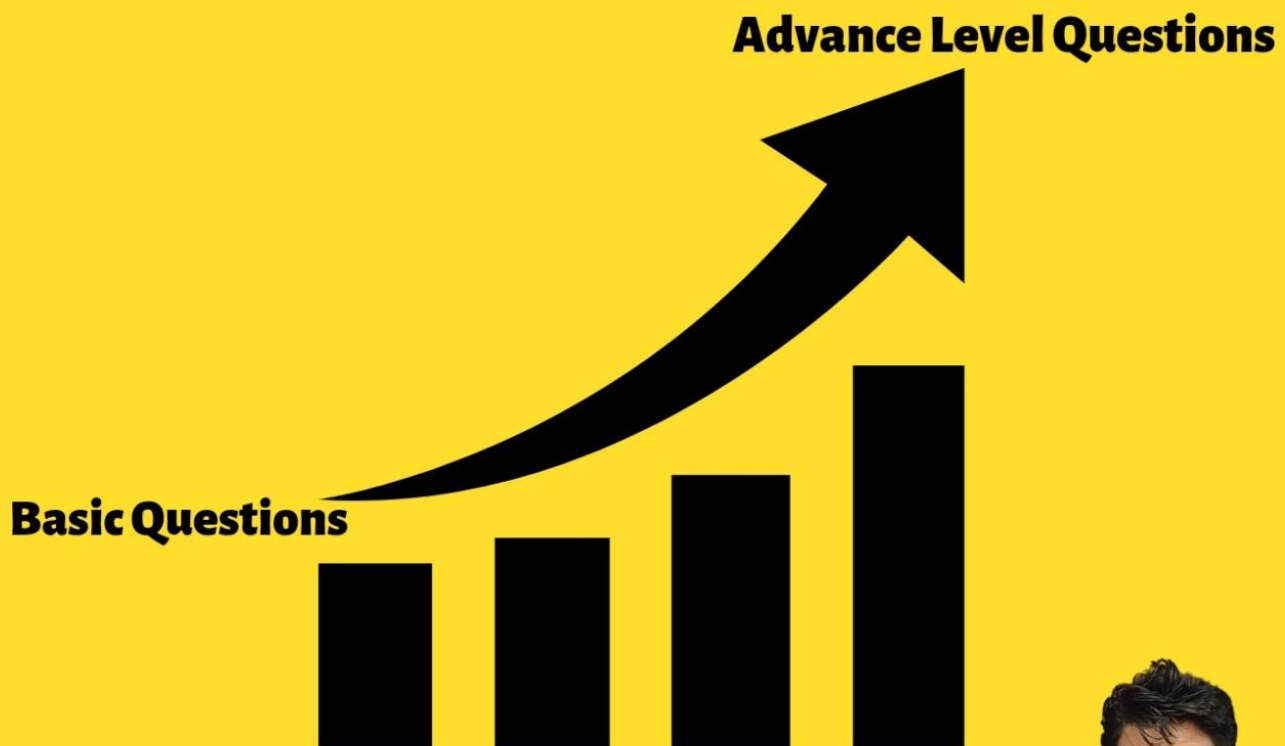

TRAIN

ALL TYPE WISE QUESTIONS

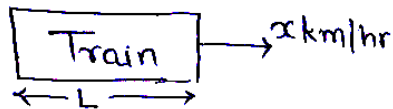


Maths with Gajanand



Train

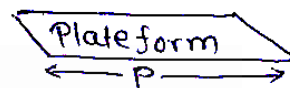
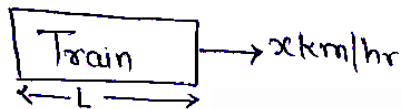
1.



↓
Pole/man

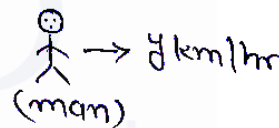
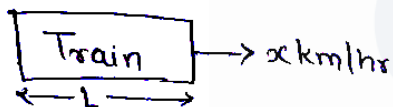
$$\left[\text{Time to cross the Pole} = \frac{L}{x} \right]$$

2.



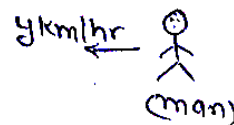
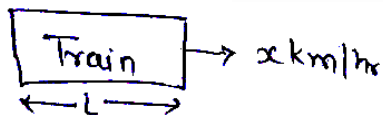
$$\left[\text{Time to Cross the Platform} = \frac{L+P}{x} \right]$$

3.



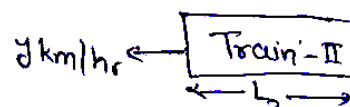
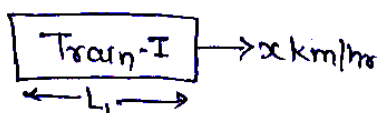
$$\left[\text{Time to cross the man} = \frac{L}{x-y} \right] \text{ if } x > y$$

4.



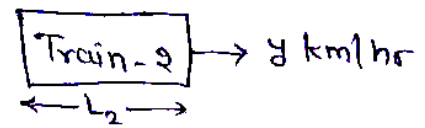
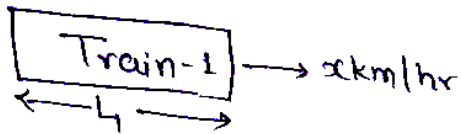
$$\left[\text{Time to cross the man} = \frac{L}{x+y} \right]$$

5.



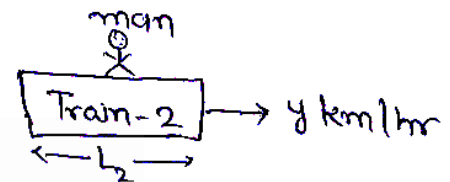
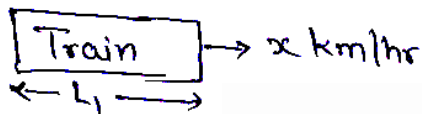
$$\left[\text{Time to cross each other} = \frac{L_1 + L_2}{x + y} \right]$$

6.



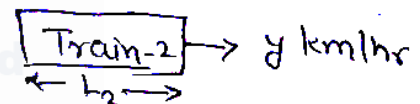
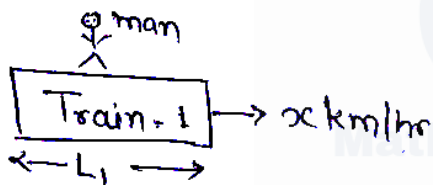
$$\left[\text{Train to cross the slower Train} = \frac{L_1 + L_2}{x - y} \right] \text{ if } x > y$$

7.



$$\left[\text{Time to cross the sitted man} = \frac{L_1}{x - y} \right] \text{ if } x > y$$

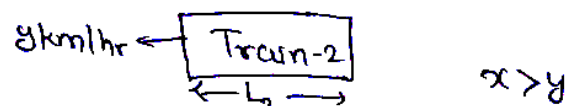
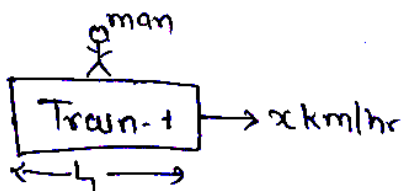
8.



$$\left[\text{Time to cross the sitted man in train-1} = \frac{L_2}{x - y} \right]$$

if $x > y$

9.



Time taken by slower train to cross a man sitting in the faster train

$$= \frac{L_2}{x + y}$$

- 1] In how much time a 280 m long train crosses a person. If speed of train is 63 km/hr.

Soln:-

$$T = \frac{280}{63 \times \frac{5}{18}} = \boxed{16 \text{ second}} \text{ Ans}$$

- 2] A train crosses an electric pole in 18 sec. If speed of train is 75 km/hr. Find the length of train.

Soln:-

$$18 = \frac{L}{75 \times \frac{5}{18}}$$

$$L = \boxed{375 \text{ m}} \text{ Ans}$$

- 3] A train, 150 m long, is running at 90 km/hr. How long (in seconds) will it take to clear a platform that is 300 m long.?

Soln:-

$$T = \frac{(150 + 300)}{90 \times \frac{5}{18}} = \boxed{18 \text{ Sec.}} \text{ Ans}$$

- 4] A train 200 m long running at 36 km/hr takes 55 seconds to cross a bridge. Length of the bridge is:

Soln:-

$$55 = \frac{(200 + l)}{36 \times \frac{5}{18}}$$

$$550 = 200 + l$$

$$l = \boxed{350 \text{ m}} \text{ Ans}$$

- [5] A train, with a uniform speed, crosses a platform, 162 metre long, in 18 seconds and another platform, 120 metre long, in 15 seconds. The speed of the train is:

Soln:- [m-I]

$$\frac{l + 162}{x} = 18$$

$$l + 162 = 18x \quad \text{--- (1)}$$

$$l = 18x - 162$$

$$\frac{l + 120}{x} = 15$$

$$l + 120 = 15x \quad \text{--- (2)}$$

$$l = 15x - 120$$

$$18x - 162 = 15x - 120$$

$$3x = 42$$

$$x = \boxed{14 \text{ m/s}}$$

$$= 14 \times \frac{18}{5} = \boxed{50.4 \text{ km/hr}} \text{ Ans}$$

[m-II] Speed of train = $\frac{\text{Difference of Platform Length}}{\text{Difference of time taken to cross the Platform}}$

$$= \frac{162 - 120}{18 - 15}$$

$$= 14 \text{ m/s} = \boxed{50.4 \text{ km/hr}} \text{ Ans}$$

- [6] A railway engine passes two bridge of lengths 400m and 235m in 100 seconds and 60 seconds, respectively. Twice the length of the railway engine (in m) is:

[CGL Pre - 2022]

Soln:-

$$\frac{x + 400}{100} = \frac{x + 235}{60}$$

$$6x + 2400 = 10x + 2350$$

$$4x = 50$$

$$2x = \boxed{25 \text{ m}} \text{ Ans}$$

- 7] A train travelling at 36 km/h crosses a pole in 25 seconds. How much time (in seconds) will it take to cross a bridge 350 m long? [SSC CGL mains-2020]

Soln:-

$$25 = \frac{l}{36 \times \frac{5}{18}}$$

$$l = 250$$

$$T = \frac{250 + 350}{36 \times \frac{5}{18}}$$

$$T = \boxed{60 \text{ second}} \quad \underline{\text{Ans}}$$

- 8] A train travelling at the speed of x km/hr crossed a 300 m long platform in 30 seconds, and overtook a man walking in the same direction at 6 km/hr in 20 seconds. What is the value of x ?

Soln:-

$$30 = \frac{l + 300}{x \times \frac{5}{18}}$$

$$20 = \frac{l}{(x-6) \times \frac{5}{18}}$$

$$100x - 600 = 18l \quad \text{--- (2)}$$

$$150x = 18l + 5400$$

$$18l = 150x - 5400 \quad \text{--- (1)}$$

from (1) & (2)

$$(1) = (2)$$

$$150x - 5400 = 100x - 600$$

$$50x = 4800$$

$$x = \boxed{96 \text{ km/hr}} \quad \underline{\text{Ans}}$$

- [9] A train passes by a lamp post at platform in 7 sec. and passes by the platform completely in 28 sec. If the length of the platform is 390m, then length of the train (in m) is:

Soln:-

$$\frac{390 + l}{x} = 28 \times \frac{5}{18} \quad \text{--- (1)}$$

$$\frac{l}{x} = 7 \times \frac{5}{18} \quad \text{--- (2)}$$

(1) - (2)

$$\frac{390}{x} = 21 \times \frac{5}{18}$$

$$x = \frac{130}{7} \text{ m/s}$$

$$l = \frac{130}{7} \times 7 = \boxed{130 \text{ m}} \text{ Ans}$$

m-II

$$\text{Speed of train} = \frac{390}{(28 - 7)}$$

$$= \frac{130}{7} \text{ m/s}$$

$$l = \frac{130}{7} \times 7$$

$$= \boxed{130 \text{ m}} \text{ Ans}$$

- [10] A passenger train 150m long is travelling with a speed of 36 km/hr. If a man is cycling in the direction of train at 9 km/hr, the time taken by the train to pass the man is:

Soln:-

$$T = \frac{l}{(x - y)}$$

$$= \frac{150}{(36 - 9) \times \frac{5}{18}}$$

$$= \frac{150}{27 \times \frac{5}{18}}$$

$$= \frac{150 \times 18}{27 \times 5}$$

$$= \boxed{20 \text{ Second}} \text{ Ans}$$

11 A train travelling at the speed of x km/h crossed a 200m long platform in 30 seconds and overtook a man walking in the same direction at the speed of 6 km/hr in 20 seconds. What is the value of x ?

Soln:-

$$\frac{200 + l}{x \times \frac{5}{18}} = 30$$

$$3600 + 18l = 150x$$

$$18l = 150x - 3600 \quad \text{--- (1)}$$

$$\frac{l}{(x-6) \times \frac{5}{18}} = 20$$

$$18l = 100x - 600 \quad \text{--- (2)}$$

$$\text{(1) = (2)}$$

$$150x - 3600 = 100x - 600$$

$$50x = 3000$$

$$x = \boxed{60} \text{ Ans}$$

12 A train 100 metre long meets a man going in opposite direction at 5 km/hr and passes him in $7\frac{1}{5}$ seconds. What is the speed of the train (in km/hr)?

Soln:-

$$T = \frac{l}{(x+y)}$$

$$\frac{36}{5} = \frac{100}{(5+y) \times \frac{5}{18}}$$

$$(5+y) \times \frac{5}{18} \times \frac{36}{5} = 100$$

$$10 + 2y = 100$$

$$2y = 90$$

$$y = \boxed{45 \text{ km/hr}} \text{ Ans}$$

m-II

$$\text{Relative speed} = \frac{100}{36/5} \times \frac{18}{5}$$

$$(5+x) = 50 \text{ km/hr}$$

$$x = \boxed{45 \text{ km/hr}}$$

Ans

13 A train of length 287 m, running at 80 km/hr, crosses another train moving in the opposite direction at 37 km/hr in 18 seconds. What is the length of the other train?

Soln.:-

[CGL-Mains
2020]

$$T = \frac{L_1 + L_2}{x + y}$$

$$18 = \frac{(287 + x)}{(80 + 37) \times \frac{5}{18}}$$

$$585 = 287 + x$$

$$x = \boxed{298 \text{ m}} \text{ Ans}$$

14 A train of length 100 m crosses another train of length 150 m, running on a parallel track in the opposite direction in 9 seconds. If the speed of train having length 150 m is 40 km/hr, then what is the speed (in km/hr) of the other train?

Soln. m-I

$$\text{Relative speed} = \frac{(100 + 150)}{9} = \frac{250}{9} \text{ m/s}$$

$$s_1 + s_2 = \frac{250}{9} \times \frac{18}{5}$$

$$40 + s_2 = 100$$

$$s_2 = \boxed{60 \text{ km/hr}} \text{ Ans}$$

m-II Basic

$$9 = \frac{(100 + 150)}{(40 + x) \times \frac{5}{18}}$$

$$(40 + x) \times \frac{5}{2} = 100 + 150$$

$$40 + x = 100$$

$$x = \boxed{60 \text{ km/hr}} \text{ Ans}$$

15 A train x running at 74 km/h crosses another train y running at 52 km/h in the opposite direction in 12 seconds. If the length of y is two-thirds that of x , then what is the length of y (in m)?

Soln:- $\frac{2}{3}$ x y
length:- $3k$ $2k$

$$12 = \frac{3k + 2k}{(52 + 74) \times \frac{5}{18}}$$

$$12 \times 126 \times \frac{5}{18} = 5k$$

$$k = 84$$

$$\text{length of } y = 2k = \boxed{168 \text{ m}} \text{ Ans}$$

16 A train 150 m long passes a stone in 30 seconds and another train of the same length travelling in opposite direction in 10 seconds. The speed of the second train is?

Soln:-

$$30 = \frac{150}{x}$$

$$x = 5 \text{ m/s}$$

$$10 = \frac{(150 + 150)}{5 + S_2}$$

$$50 + 10S_2 = 300$$

$$S_2 = 25 \text{ m/s}$$

$$= 25 \times \frac{18}{5}$$

$$= \boxed{90 \text{ km/hr}} \text{ Ans}$$

17 Two trains each having a length of 160 meters moving in opposite direction crossed each other in 9 seconds. If one train crossed a 200-metre-long platform in 27 seconds, then the ratio of their speeds is:

Soln:-

$$9 = \frac{(160+160)}{x+y}$$

$$x+y = \frac{320}{9}$$

$$y = \frac{320}{9} - \frac{40}{3}$$

$$y = \frac{200}{9}$$

$$27 = \frac{(200+160)}{x}$$

$$x = \frac{360}{27} = \frac{40}{3} \text{ m/s}$$

$$x:y = \frac{40}{3} : \frac{200}{9} = \boxed{3:5} \text{ Ans}$$

18 A train travelling at 48 km/hr crosses another train, having half its length and travelling in opposite direction at 42 km/hr, in 12 seconds. It also passes a railway platform in 45 seconds. The length of the railway platform is:

Soln:-

$$12 = \frac{l + \frac{l}{2}}{\frac{(48+42) \times 5}{18}}$$

$$300 = \frac{3l}{2}$$

$$l = 200 \text{ m}$$

Let Platform length = P

$$45 = \frac{200+P}{\frac{48 \times 5}{18}}$$

$$600 = 200 + P$$

$$P = \boxed{400 \text{ m}} \text{ Ans}$$

19 How much time does a train 50m long, moving at 68 km/hr takes to pass another train 75m long moving at 50 km/hr in the same direction?

Soln:

$$T = \frac{L_1 + L_2}{x - y}$$

$$T = \frac{(50 + 75)}{(68 - 50) \times \frac{5}{18}}$$

$$= \frac{125}{18 \times \frac{5}{18}} = \boxed{25 \text{ Second}} \text{ Ans}$$

20 A train 'B' speeding with 100 km/hr crosses another train C, running in the same direction in 2 mins. If the length of the train B and C be 150m and 250m respectively, what is the speed of the train C (in km/hr)

Soln: m-I

$$2 \times 60 = \frac{(150 + 250)}{(100 - x) \times \frac{5}{18}}$$

$$120 \times (100 - x) \times \frac{5}{18} = 400$$

$$600 - 6x = 72$$

$$100 - x = 12$$

$$x = \boxed{88 \text{ km/hr}} \text{ Ans}$$

m-II

Distance = Relative speed
× Time

$$\frac{(150 + 250)}{1000} = (100 - x) \times \frac{1}{30} \text{ hr}$$

$$0.4 = (100 - x) \times \frac{1}{30}$$

$$x = \boxed{88 \text{ km/hr}} \text{ Ans}$$

Q1 Two trains are running on a Parallel line in the same direction at a speed of 50 km/h and 30 km/hr respectively. Given that the faster train crosses a man in the slower train in 18 sec, what is the length of the faster train?

Soln:-

$$T = \frac{L}{x-y}$$

$$18 = \frac{L}{(50-30) \times \frac{5}{18}}$$

$$L = \boxed{100m} \text{ Ans}$$

OR

$$\text{Length} = \text{Relative speed} \times \text{Time}$$

$$= (50-30) \times \frac{5}{18} \times 18$$

$$= \boxed{100m} \text{ Ans}$$

Q2 Two trains are moving in the opposite direction at speed of 43 km/hr and 51 km/hr respectively. The time taken by the slower train to cross a man sitting in the faster train is 9 seconds. What is the length (in metres) of the slower train?

Soln:-

$$T = \frac{L}{x+y}$$

$$9 = \frac{L}{(43+51) \times \frac{5}{18}}$$

$$9 \times 94 \times \frac{5}{18} = L$$

$$L = \boxed{235m} \text{ Ans}$$

23 Renu was sitting inside train A, which was travelling at 50 km/h. Another train, B whose length was three times the length of A crossed her in the opposite direction in 15 seconds. If the speed of train B was 58 km/h then the length of train A (in m) is:

Soln:-

length of	A	B
	l	$3l$

$$15 = \frac{3l}{(50+58) \times \frac{5}{18}}$$

$$15 \times 5 \times 108 = 3l \times 18$$

$$l = \boxed{150m} \text{ Ans}$$

24 Two trains start at same time from stations A to B, 1800 km apart, and proceed towards each other at an average speed of 44 km/h and 46 km/h respectively, where will the train meets? [CHSL-2020]

Soln:-

$$T = \frac{1800}{(44+46)}$$

$$= \frac{1800}{90} = 20 \text{ hour}$$

$$20 \times 44 \text{ km/hr} = \boxed{880 \text{ km}} \text{ from station A}$$

OR

$$20 \times 46 \text{ km/hr} = 920 \text{ km from station B}$$

Q5 Two trains start from a certain place on two parallel tracks in the same direction. The speed of the trains are 45 km/hr and 40 km/hr respectively. The distance between the two trains after 45 minutes will be.

Soln:-

$$D = \text{Relative speed} \times \text{Time}$$

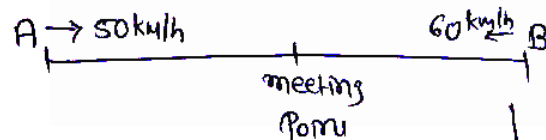
$$= (45 - 40) \times \frac{45}{60}$$

$$= \frac{15}{4}$$

$$= \boxed{3.75 \text{ km}} \text{ Ans}$$

Q6 Two trains start from station A and B and travel towards each other at speed of 50 km/hr and 60 km/hr respectively. At the time of their meeting, the second train has travelled 120 km more than the first. The distance between A and B is:

Soln:-



T^x

$$\text{Speed} \rightarrow 50 : 60$$

$$5 : 6$$

$$\text{Distance} \rightarrow 5 : 6$$

$$\frac{1}{1} \rightarrow 120 \text{ km}$$

$$\text{Distance btw A to B} = (5+6) \text{ Unit}$$

$$= 11 \times 120$$

$$= \boxed{1320 \text{ km}} \text{ Ans}$$

Time taken by trains
m-II before meeting points

$$= \frac{120 \text{ km}}{(60 - 50) \text{ km/hr}}$$

$$= 12 \text{ hr}$$

$$D = (60 + 50) \times 12$$

$$= \boxed{1320 \text{ km}}$$

Ans

Q7 A train without stoppage travels with an average speed of 65 km/h and with stoppage, it travels with an average speed of 52 km/hr. For how many minutes does the train stop on an average per hour? [CGL-2019]

Soln:-

In 1 hr $\rightarrow$ 65 km

In 1 hr $\rightarrow$ 52 km } 13 km less

$$T = \frac{13}{65} \times 60 = \boxed{12 \text{ min}}$$

Ans

m-II

$$\left[\frac{\text{Faster speed} - \text{Slower speed}}{\text{Faster speed}} \times 60 \right]$$

$$= \frac{65 - 52}{65} \times 60 \text{ min/hr}$$

$$= \boxed{12 \text{ min}} \text{ Ans}$$

Q8 A train crosses a man going along the railway track at 13.5 km/hr. The man could see the train upto 4 min then find the speed of the train if due to fog he can see upto 1.9 km and length of train is 500 metre?

Soln:-

$$4 \times 60 = \frac{500 + 1900 \text{ m}}{(S - 13.5) \times \frac{5}{18}}$$

$$240(S - 13.5) \times \frac{5}{18} = 2400$$

$$S - 13.5 = 36$$

$$S = \boxed{49.5 \text{ km/hr}} \text{ Ans}$$

29 A train crosses a person moving in the same direction with a speed of 18 km/hr. If person can see the train for 2.5 min. If person can see only 1700 m due to fog. Find the length of train if speed of train is 72 km/hr.

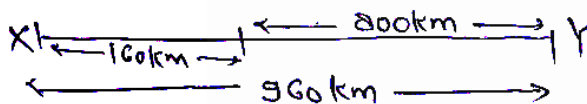
Soln:-

$$\begin{aligned} D &= \text{Relative speed} \times \text{Time} \\ &= (72 - 18) \times \frac{5}{18} \times 2.5 \times 60 \\ &= 2250 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{Length of train} &= 2250 - 1700 \\ &= \boxed{550 \text{ m}} \text{ Ans} \end{aligned}$$

30 The distance between 2 stations X and Y is 960 km. If 2 trains start together at same time from both stations towards each other and meet after 12 hours but if one train is started 6 hours 24 minutes after the first then they meet after 10 hours. Find the speed of trains?

Soln:-



$$S_1 + S_2 = \frac{960}{12} = 80 \text{ km/hr}$$

$$\text{In 10 hrs} = 80 \times 10 = 800 \text{ km}$$

$$\text{Remaining} = 960 - 800 = 160 \text{ km} \xrightarrow{\text{travel}} \text{in 6 hr 24 min}$$

$$\text{Speed} = \frac{160}{32/5} = \boxed{25 \text{ km/hr}}$$

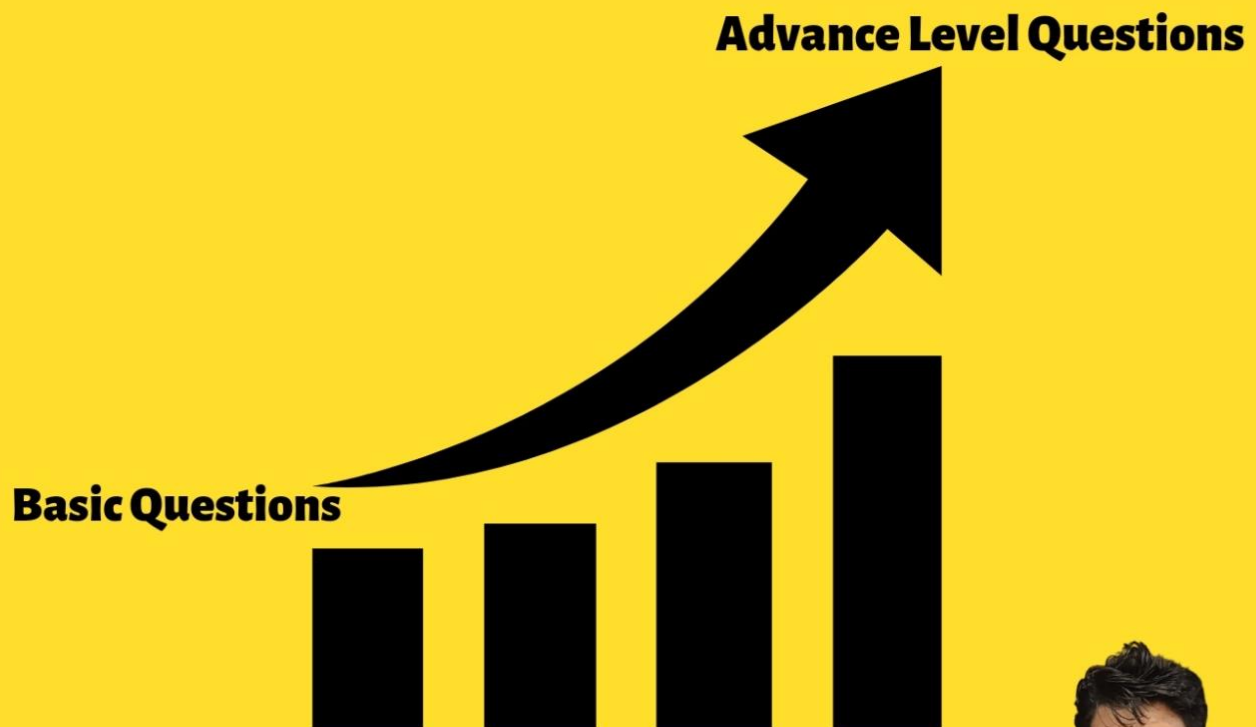
$$\begin{aligned} S_1 + S_2 &= 80 \\ \downarrow \\ 25 \end{aligned}$$

$$S_2 = \boxed{55 \text{ km/hr}}$$

Ans

BOAT & STREAM

ALL TYPE WISE QUESTIONS



Maths with Gajanand

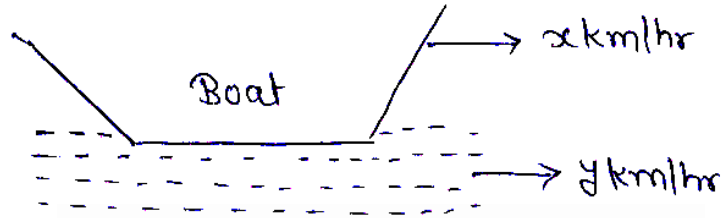


Boat & Stream

1. Speed of Boat in still water $= x \text{ km/hr}$

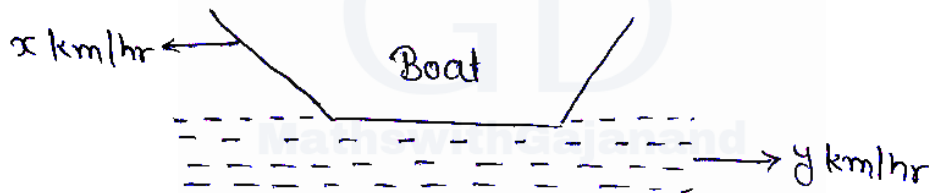
Speed of current/stream $= y \text{ km/hr}$

(i)



Speed of Boat in down stream $= (x+y) \text{ km/hr}$

(ii)



Speed of Boat in upstream $= (x-y) \text{ km/hr}$ $x > y$

2. If Speed of Boat in downstream $= a \text{ km/hr}$

and Speed of Boat in upstream $= b \text{ km/hr}$

then

Speed of Boat in still water $= \frac{a+b}{2} \text{ km/hr}$

Speed of stream/current $= \frac{a-b}{2} \text{ km/hr}$

- 1 The speed of a boat in still water is 10 km/hr. It covers (upstream) a distance of 45 km in 6 hours. The speed (in km/hr) of the stream is :

Soln:-

$$x = 10 \text{ km/hr}$$

$$x - y = \frac{45}{6}$$

$$y = 10 - \frac{45}{6}$$

$$y = \frac{15}{6}$$

$$y = \boxed{2.5 \text{ km/hr}} \text{ Ans}$$

- 2 A man swims downstream distance of 15 km in 1 hour. If the speed of the current is 5 km/hr, the time taken by the man to swim the same distance upstream is :

Soln:-

$$x + y = \frac{15}{1}$$

$$y = 5 \text{ km/hr}$$

$$x = 15 - 5 = 10 \text{ km/hr}$$

$$\begin{aligned} x - y &= 10 - 5 \\ &= 5 \text{ km/hr} \end{aligned}$$

$$\text{time} = \frac{15 \text{ km}}{5 \text{ km/hr}}$$

$$= \boxed{3 \text{ hour}} \text{ Ans}$$

- [3] A man rows 12 km in 5 hours against the stream and the speed of current being 4 km/hr. What time will be taken by him to row 15 km with the stream?

Soln:-

$$x - y = \frac{12}{5}, \quad y = 4 \text{ km/hr}$$

$$x = 4 + \frac{12}{5}$$

$$x = \frac{32}{5} = 6.4 \text{ km/hr}$$

$$x + y = 6.4 + 4 = 10.4 \text{ km/hr}$$

$$\text{Time} = \frac{15}{10.4} = \frac{150}{104}$$

$$= \boxed{1 \text{ hour } 26 \frac{7}{13} \text{ minutes}} \quad \underline{\underline{\text{Ans}}}$$

- [4] The speed of a boat in still water is 15 km/hr, and the speed of the current is 5 km/hr. In how much time (in hour) will the boat travel a distance of 60 km upstream and the same distance downstream?

Soln:-

$$x = 15 \text{ km/hr}$$

$$y = 5 \text{ km/hr}$$

$$\text{Time} = \frac{60}{15-5} + \frac{60}{15+5}$$

$$= 6 + 3$$

$$= \boxed{9 \text{ hr}} \quad \underline{\underline{\text{Ans}}}$$

- [5] Speed of motorboat in still water is 45 km/hr. If the motorboat travels 80 km along the stream in 1 hour 20 minutes, then the time taken by it to cover the same distance against the stream will be:-

Soln:-

$$x = 45 \text{ km/hr}$$

$$x+y = \frac{80}{1+\frac{20}{60}} = \frac{80}{1+\frac{1}{3}} = \frac{80 \times 3}{4}$$

$$x+y = 60 \text{ km/hr}$$

$$y = 60 - 45$$

$$y = 15 \text{ km/hr}$$

$$x-y = 45 - 15 = 30 \text{ km/hr}$$

$$\text{Time} = \frac{80}{30} = \frac{8}{3} = \boxed{2 \text{ hour } 40 \text{ min}} \quad \underline{\underline{\text{Ans}}}$$

- [6] A motorboat goes 24 km in 2 hour along the stream and 10 km in 1 hour against the stream. The speed of the motorboat in kilometres per hour, is

Soln:-

$$a = x+y = \frac{24}{2} = 12 \text{ km/hr}$$

$$b = x-y = \frac{10}{1} = 10 \text{ km/hr}$$

$$x = \frac{a+b}{2} \\ = \frac{12+10}{2}$$

$$x = \boxed{11 \text{ km/hr}} \quad \underline{\underline{\text{Ans}}}$$

7] A boat travels 24 km upstream in 6 hours and 20 km downstream in 4 hours. Then the speed of boat in still water and the speed of current are respectively?

Soln:-

$$x - y = a = \frac{24}{6} = 4 \text{ km/hr}$$

$$x + y = b = \frac{20}{4} = 5 \text{ km/hr}$$

$$x = \frac{5+4}{2}, \quad y = \frac{5-4}{2}$$

$$= \boxed{4.5 \text{ km/hr}}_{\text{Ans}} \quad = \boxed{0.5 \text{ km/hr}}_{\text{Ans}}$$

8] A man can row 6 km/hr in still water. If the speed of the current is 2 km/hr, he takes 4 hours more in upstream than in the downstream. The distance is:

Soln:-

$$x = 6 \text{ km/hr}, \quad y = 2 \text{ km/hr}$$

$$\frac{D}{x-y} = \frac{D}{x+y} + 4 \text{ hr}$$

$$\frac{D}{4} = \frac{D}{8} + 4$$

$$\frac{D}{8} = 4$$

$$D = \boxed{32 \text{ km}}_{\text{Ans}}$$

m-II

Speed $\rightarrow$ Downstream : Upstream
8 km/hr : 4 km/hr = 2 : 1

Time $\rightarrow$ 1 : 2
4 hr : 8 hr

$$\text{Distance} = 8 \text{ km/hr} \times 4$$

$$= \boxed{32 \text{ km}}_{\text{Ans}}$$

9] A man goes downstream with a boat to some destination and return upstream to his original place in 5 hours. If the speed of the boat in still water and the stream are 10 km/hr and 4 km/hr respectively, the distance of the destination from the starting place is:

Soln:- m-I
Basic

$$x = 10 \text{ km/hr}$$

$$y = 4 \text{ km/hr}$$

$$x+y = 14 \text{ km/hr} \quad , \quad x-y = 6 \text{ km/hr}$$

$$\frac{D}{14} + \frac{D}{6} = 5 \text{ hr}$$

$$\frac{10D}{42} = 5$$

$$D = \boxed{21 \text{ km}}$$

m-II

$$D = \frac{(x^2 - y^2)}{2x} \times T$$

$$= \frac{(10^2 - 4^2)}{2 \times 10} \times 5$$

$$= \frac{84 \times 5}{20}$$

$$D = \boxed{21 \text{ km}} \quad \underline{\underline{\text{Ans}}}$$

- 10** Speed of a boat is 5 km/hr in still water. the Speed of the stream is 3 km/hr. If the boat takes 3 hr to go to a place and come back, the distance of the place is:

Soln → **m-1**

$$\frac{D}{5+3} + \frac{D}{5-3} = 3 \text{ hr}$$

$$\frac{D}{8} + \frac{D}{2} = 3 \text{ hr}$$

$$\frac{5D}{8} = 3 \text{ hr}$$

$$D = \frac{24}{5}$$

$$= \boxed{4.8 \text{ km}} \text{ Any}$$

m-11

$$D = \frac{x^2 - y^2}{2x} \times T$$

$$= \frac{25 - 9}{2 \times 5} \times 3$$

$$D = \boxed{4.8 \text{ km}} \text{ Any}$$

- 11** A boat goes 12 km downstream and comes back to the starting point in 3 hours. If the speed of the current is 3 km/hr, then the speed (in km/hr) of the boat in still water is ?

Soln

$$\frac{12}{x+y} + \frac{12}{x-y} = 3$$

$$\frac{12}{x+3} + \frac{12}{x-3} = 3$$

$$\frac{12(2x)}{x^2-9} = 3$$

$$8x = x^2 - 9$$

$$x^2 - 8x - 9 = 0$$

$$x = 9, -1$$

$$\boxed{9 \text{ km/hr}} \text{ Any}$$

$$D = \frac{x^2 - y^2}{2x} \times T$$

$$12 = \frac{x^2 - 3^2}{2x} \times 3$$

$$x^2 - 8x - 9 = 0$$

$$x = 9, -1$$

$$\boxed{9 \text{ km/hr}} \text{ Any}$$

12 The speed of the current is 5 km/hour. A motorboat goes 10 km upstream and back again to the starting point in 50 minutes. The speed (in km/hr) of the motorboat in still water is:

Soln:- (a) 20 (b) 26 ~~(c) 25~~ (d) 28

$$\frac{10}{x+5} + \frac{10}{x-5} = \frac{5}{6}$$

by optim:- check $x=25$

$$\frac{10}{30} + \frac{10}{20} \Rightarrow \frac{5}{6} \text{ satisfy } \checkmark$$

13 A man can row 30 km downstream and return in a total of 8 hours. If the speed of the boat in still water is four times the speed of the current, then the speed of the current is:

Soln:- $x = 4y$

$$\frac{30}{x+y} + \frac{30}{x-y} = 8$$

$$\frac{30}{5y} + \frac{30}{3y} = 8$$

$$y = \boxed{2 \text{ km/hr}} \text{ Ans}$$

m-II

$$D = \frac{x^2 - y^2}{2x} \times T$$

$$30 = \frac{(4y)^2 - y^2}{2 \times 4y} \times 8$$

$$30 = \frac{15y^2}{8y} \times 8$$

$$y = \boxed{2 \text{ km/hr}} \text{ Ans}$$

14 A man rows to a place 35 km in distance and back in 10 hours 30 minutes. He found that he can row 5 km with the stream in the same time as he can row 4 km against the stream. Find the rate of flow of the stream?

Soln:-

$$\frac{5}{x+y} = \frac{4}{x-y}$$

$$5x - 5y = 4x + 4y$$

$$x = 9y$$

$$\frac{x}{9} = \frac{9}{4} = \frac{9k}{k}$$

$$\frac{35}{10k} + \frac{35}{8k} = \frac{21}{2}$$

$$\frac{35}{2k} \left[\frac{1}{5} + \frac{1}{4} \right] = \frac{21}{2}$$

$$\frac{5}{k} \times \frac{9}{20} = 3$$

$$k = \frac{3}{4} = 0.75$$

$$y = k = \boxed{0.75 \text{ km/hr}} \text{ Ans}$$

[15] If a boat goes a certain distance at 30 km/hr and comes back the same distance at 60 km/hr. What is the average speed (in km/hr) for the total journey?

Soln:-

$$\text{avg. speed} = \frac{2xy}{x+y}$$

$$= \frac{2 \times 30 \times 60}{30 + 60}$$

$$= \frac{2 \times 30 \times 60}{90}$$

$$= \boxed{40 \text{ km/hr}} \text{ Ans}$$

- 16] A boat can cover a distance of 56 km downstream in 3.5 hours. The ratio of the boat in still water and the speed of stream is 3:1. How much time (in hours) will the boat take to cover a distance of 41.6 km downstream?

Soln:- speed of boat in downstream = $\frac{56}{3.5}$
 $= 16 \text{ km/hr}$

Required time = $\frac{41.6}{16} = \boxed{2.6 \text{ hrs}}$ Ans.

- 17] Speed of a boat in still water is 6 km/hr. Time taken by the boat in upstream is four times the time taken in downstream. Then find the speed of the current.

Soln:-

$$4 \left(\frac{D}{6+y} \right) = \frac{D}{6-y}$$

$y = \boxed{3.6 \text{ km/hr}}$ Ans

- 18] A boat takes half time in moving a certain distance downstream than upstream. The ratio of the speed of the boat in still water and that of the current is:

Soln:- $\frac{D}{x+y} = \frac{1}{2} \frac{D}{(x-y)}$

$$2x - 2y = x + y$$

$$x = 3y$$

$$\frac{x}{y} = \frac{3}{1}$$

$\boxed{3:1}$ Ans

Ans II $\begin{array}{l} \text{Downstream} \\ T \rightarrow 1 : 2 \\ \text{Upstream} \\ S \rightarrow 2 : 1 \end{array}$

$$x = \frac{2+1}{2} = \frac{3}{2}$$

$$y = \frac{2-1}{2} = \frac{1}{2}$$

$x:y = \frac{3}{2} : \frac{1}{2} = \boxed{3:1}$ Ans

19 A motor boat covers a certain distance downstream in a river in 3 hours. It covers the same distance upstream in three and half an hour. If the speed of the water is 1.5 km/hr, then the speed of the boat in still water is:-

Soln:-

$$\frac{D}{x+1.5} = 3 \text{ hr}$$

$$\frac{D}{x-1.5} = 3.5 \text{ hr}$$

$$D = 3(x+1.5) \text{ --- (1)}$$

$$D = \frac{7}{2}(x-1.5) \text{ --- (2)}$$

$$(1) = (2) \quad 3(x+1.5) = \frac{7}{2}(x-1.5)$$

$$6x + 9 = 7x - 10.5$$

$$x = \boxed{19.5 \text{ km/hr}} \text{ Ans}$$

20 A boat goes 15 km upstream and 22 km downstream in 5 hours. It goes 20 km upstream and $\frac{55}{2}$ km downstream in $\frac{13}{2}$ hours. What is the speed (in km/hr) of stream?

Soln:-

$$\frac{15}{x-y} + \frac{22}{x+y} = 5$$

$$\frac{20}{x-y} + \frac{55}{2(x+y)} = \frac{13}{2}$$

$$\frac{1}{x-y} = X$$

$$15X + 22Y = 5 \text{ --- (1)}$$

$$20X + \frac{55}{2}Y = \frac{13}{2} \text{ --- (2)}$$

$$\frac{1}{x+y} = Y$$

$$(1) \times 4 - (2) \times 3$$

$$Y = \frac{1}{11}, \quad X = \frac{1}{5}$$

$$x+y = 11$$

$$x-y = 5$$

$$x = \frac{16}{2} = 8$$

$$y = \frac{6}{2} = \boxed{3 \text{ km/hr}} \text{ Ans}$$

Q1 A boat can go 20km downstream and 30km upstream in 2 hours 20 minutes. Also, it can go 10km downstream and 8km upstream in 49 minutes. What is the speed of boat downstream in km/hr?

Soln:-

$$\frac{20}{x+y} + \frac{30}{x-y} = 2 + \frac{20}{60} \quad \text{--- ①}$$

$$\frac{10}{x+y} + \frac{8}{x-y} = \frac{49}{60} \quad \text{--- ②}$$

$$\text{②} \times 15 - \text{①} \times 4$$

$$\frac{150}{x+y} - \frac{80}{x-y} = \frac{49}{4} - \frac{28}{3}$$

$$\frac{70}{x+y} = 7 \left(\frac{5}{12} \right)$$

$$x+y = \boxed{24 \text{ km/hr}} \text{ Ans}$$

Q2 The water in a river is flowing at the rate of 4 km/hr. If the width and depth of the river is 8m and 4m respectively, then how much water will enter the sea in 15 minutes?

Soln:- If water flows at 4 km/hr

$$\text{So in 15 min, it travels} = 4 \times \frac{15}{60} = 1 \text{ km}$$

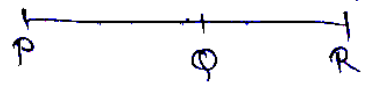
$$\text{Volume} = l b h$$

$$= 1000 \text{ m} \times 8 \text{ m} \times 4 \text{ m}$$

$$= \boxed{32000 \text{ m}^3} \text{ Ans}$$

Q3 On a river, Q is the mid-point between two points P and R on the same bank of the river. A boat can go from P to Q and back in 12 hours, and from P to R in 16 hours 40 min. How long would it take to go from R to P?

Soln:- Time required travelled from



$$P \text{ to } R = 16 \text{ hr } 40 \text{ min}$$

$$\begin{aligned} \text{So } P \text{ to } Q &= \frac{1}{2}(16 \text{ hr } 40 \text{ min}) \\ &= 8 \text{ hr } 20 \text{ min} \end{aligned}$$

$$\text{Total required time from } P \text{ to } Q + Q \text{ to } P = 12 \text{ hr}$$

$$Q \text{ to } P = 12 - 8 \text{ hr } 20 \text{ min}$$

$$\overrightarrow{QP} = 3 \text{ hr } 40 \text{ min.}$$

$$\text{Required time to } \overrightarrow{RP} = 2 \times \overrightarrow{QP}$$

$$= 2[3 \text{ hr } 40 \text{ min}]$$

$$= 7 \text{ hr } 20 \text{ min} = \boxed{7\frac{1}{3} \text{ hr}}$$

Ans

Q4 Two boat are travelling with speed of 36 km/hr and 54 km/hr respectively towards each other. What is the distance (in metres) between the two boats one second before they collide?

[SSC CGL-2020]

Soln:-

$$\text{Distance} = \text{Relative Speed} \times \text{Time}$$

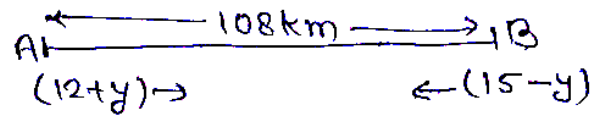
$$= (36 + 54) \times \frac{5}{18} \times 1 \text{ sec}$$

$$= 90 \times \frac{5}{18} \times 1$$

$$= \boxed{25 \text{ metre}} \text{ Ans}$$

Q5 Two boats A and B start towards each other from two places, 108 km apart. Speed of the boat A and B in still water are 12 km/hr and 15 km/hr respectively. If A proceeds down and B up the stream, they will meet after.

Soln:-



Relative speed of A and B

$$= (12+y) + (15-y)$$

$$= 12+15 = 27 \text{ km/hr}$$

$$\text{Time} = \frac{\text{Distance}}{\text{Relative speed}} = \frac{108}{27} = \boxed{4 \text{ hour}} \text{ Ans}$$

Q6 A swimmer swims from a point against a current for 5 minutes and then swims backwards in favour of the current for next 5 minutes and comes to the points B. If AB is 100 metres, the speed of the Current (in km/hr) is:-

Soln:-

$$OA = (x-y) \times \frac{5}{60}$$

$$OB = (x+y) \times \frac{5}{60}$$

$$AB = OB - OA$$

$$= (x+y) \times \frac{5}{60} - (x-y) \times \frac{5}{60} = \frac{100}{1000} \text{ km}$$

$$\frac{5}{60} [x+y - x+y] = \frac{1}{10}$$

$$y = \boxed{0.6 \text{ km/hr}} \text{ Ans}$$

